

Embedding Agentic Traces

A trace representation should be faithful to content and invariant to authorship

Helivan Research — working summary, August 2026

The problem

Modern generative systems produce logs of their actions and decisions—referred to as an *agentic trace*—while processing information. Traces often implicitly or explicitly encode information such as the particular type of tool that was used, the harness, etc. that may or may not be useful for determining if the trace produced a useful output. Further, the signal from using an off-the-shelf embedding function to encode trace similarity is often dominated by information that is potentially irrelevant for correctness, such as the harness or model used: in our corpus, nearest neighbors of raw trace embeddings are same-system 68–83% of the time, same-scaffold at $5\times$ chance, and the exact agent configuration is retrievable at 0.82 vs. 0.43 chance even among same-model, same-harness siblings—while agreement with outcome sits at chance (AUC 0.52).

The goal of this work is simple: **develop a method for embedding agentic traces that better captures similarity of the work performed.** In developing such a method, we have identified six desirable properties of an agentic trace embedding, organized into three groups:

Faithful to content — what happened should drive similarity.

- **Task fidelity:** traces of different systems working the same problem should be similar. *Measured by* cross-system same-task nearest-neighbor retrieval.
- **Behavioral fidelity:** among traces of the same problem, similar approaches and results should be similar. *Measured by* within-task coherence of same-outcome traces (AUC of same- vs. different-outcome distances).

Invariant to authorship — who produced it should not.

- **Harness invariance:** scaffold formatting and tool idiom should not drive similarity. *Measured by* within-task same-scaffold retrieval vs. chance.
- **Model-family invariance:** the vendor/lineage of the underlying LLM should not drive similarity. *Measured by* within-task same-family retrieval vs. chance.
- **Identity invariance:** the exact system configuration (version, prompts, settings) should not be recoverable. *Measured by* exact-submission retrieval among same-model, same-harness siblings vs. chance.

Reliable — similarity should be a measurement, not noise.

- **Stability:** the embedding should be stable under agent stochasticity, judge resampling, and embedder choice. *Measured by* replicate-run and judge-resample agreement, per trace and aggregated over instances.

No off-the-shelf representation satisfies these: a dozen

unsupervised variants of the naive representation are performance-identical (an information ceiling), and geometric fixes backfire—per-instance centering alone *amplifies* provenance dominance ($0.68 \rightarrow 0.83$).

qubric: query-specific rubrics

For each task instance, an LLM reads the *public problem statement* (no labels) and writes a query-specific rubric: what understanding, localization, reproduction, editing, verification, and final state concretely mean for *this* problem. A judge LLM reads each trace through the rubric and extracts a short factual description per section; sections are embedded separately, centered against the cross-system median per instance (consensus centering), and concatenated. The rubric is a natural-language definition of the equivalence relation the space should respect; invariance comes from *re-observation through a fingerprint-poor channel*, not surgery on a fingerprint-rich one.

Evidence, by property

Across **seven embedders** (three families, two orders of magnitude of size) qubric produces the same profile shift (Fig. 1): *task fidelity* is retained (~ 0.95) and *behavioral fidelity* retained or improved, while every authorship axis collapses to $\approx$ chance—*harness* $0.31 \rightarrow 0.08$, *model family* $0.20 \rightarrow 0.12$, *identity* $0.83 \rightarrow 0.09$. The properties are separately measurable (double dissociations across six representations), so these are genuinely different axes being moved, not one metric relabeled.

The instance-conditioning is the active ingredient: the construction ranking is strict under both readouts (qubric > verdict questions > generic rubric > free-form summary > raw), reproduces under a local embedder, and no single rubric section carries the effect (best single section 0.091 vs. 0.078 for the union). Judge choice degrades gracefully: an open frontier judge (DeepSeek-v3.1, 0.083/0.079) matches the closed reference (gpt-5.4-mini, 0.078/0.078) within panel resolution, and a weak judge *with* a qubric outperforms a strong judge without one.

Stability: two judges agree on a single trace only 12–28% of the time, but aggregating ~ 20 instances drives system-level agreement to 0.82–0.95; replicate runs of the same agent retrieve each other at $6\times$ chance. Per-trace noisy, per-agent reliable—and per-agent is what evaluation uses.

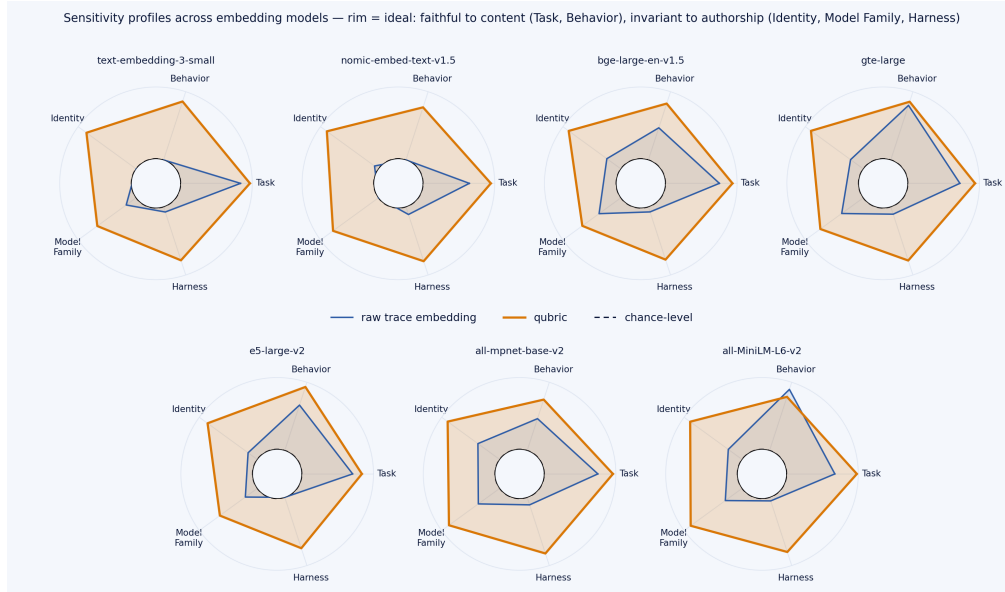


Figure 1: Sensitivity radars across seven embedders. Outside = desirable. Raw embeddings (navy) are authorship-dominated; qubric (amber) collapses Identity / Model family / Harness to chance while holding Task and Behavior.

Payoff: query-efficient benchmarking

Qubric representations enter the DKPS/PKPS perspective space; a new agent’s full-benchmark resolve rate is predicted from a few probe traces against references’ cached coverage, blended with the sample score. Evaluation is leave-one-LLM-out throughout (naive CV inflates accuracy $\sim 25\%$ via model-family leakage).

probes m	sample score	qubric pipeline	+selection
1	0.44	0.095	—
5	0.16	0.074	0.057
20	0.06	0.048–0.053	0.053

Table 1: MAE predicting true resolve rates (107 systems). One probe trace ≈ 8 –10 scored runs.

One trace carries far more than one bit: the strongest

correctness-only competitor (CAPA-style error overlap) needs 15–20 *scored* probes to match one *trace* (0.37 MAE at $m=1$), and trails the ensemble at every budget. The reference pool, not probe count, is the binding resource: error falls $0.145 \rightarrow 0.078$ as the pool grows $5 \rightarrow 96$, still declining (Fig. 2). Every layer is ablated; CV-inside-selection and PCA-composed-with-selection are documented overfitting traps.

Status & caveats

All qubric numbers: 20-instance panel of SWE-bench Verified, one benchmark; differences < 0.01 near panel resolution. The 150-instance scale-up is specified and unblocked (open-judge stack validated via OpenRouter; $\sim \$40$ –85). Terminal-Bench (142 systems, 89 tasks, ≥ 5 trials/task, public trajectories) is scoped and deferred. Full log: FINDINGS.md (F1–F24) on the **agentic-traces** branch.



Figure 2: QUENCH: error predicting true resolve rates vs. probe budget m (left) and vs. reference-pool size n (right); leave-one-LLM-out, 107 systems.